Tutorial 3 Solutions

Exercise 1

- **a. Degree = 5.**
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- **b. Possible Domains:**
  - `vehicle_id`: Strings composed of a fixed 3-letter classification prefix (`TRK` or `VAN`) followed by a hyphen and a 3-digit numerical identifier code.
  - `primary_driver`: Text strings representing the last name of the employee, followed by a comma, a space, and the initial letter of their first name completed with a period (e.g., `"Anderson, T."`).
  - `capacity_tons`: The set of strict positive integers representing structural cargo load capacity boundaries (e.g., values (>0)).

Exercise 2

- **a. Vehicle Table Correction:** Isolate driver information into its own entity to eliminate update and insertion anomalies.
  - `Vehicle (vehicle_id, model_name, capacity_tons, driver_id, last_maintenance)`
  - `Driver (driver_id, Last_Name, First_Name_Initial)`
- **b. Medical_Clinics Table Correction:** Atomic splitting of combined geographical data elements and structural person components.
  - `Clinics (Clinic_ID, Clinic_Name, City, State, Physician_ID)`
  - `Physicians (Physician_ID, First_Name, Last_Name)`
- **c. Corporate_Assets Table Correction:** Cleanse currency attributes to hold homogeneous unit calculations.
  - `Assets (Asset_ID, Asset_Name, Description, Unit_Value_USD)`
- **d. Academic_Research Table Correction:** Resolve the non-atomic, multi-valued field array (`Co_Authors_Names`) by breaking the many-to-many relationship into an intermediate bridge entity.
  - `Research_Paper (DOI_Number, Paper_Title, Publication_Year)`
  - `Author (Author_ID, Author_First_Name, Author_Last_Name)`
  - `Paper_Author_Bridge (DOI_Number, Author_ID)`

Exercise 3

Data Type	Example Real-World Domain
String of characters	`Customer_Email_Address` or `Workout_Name`
Integer	`Available_Class_Capacity` or `Quantity of an item`
Decimal number	`Monthly_Membership_Fee` or `Price of a product`
Date	`Membership_Enrollment_Date` or `Birthdate`
Boolean	`Is_Active_Member` or `Has_Paid_Fees`

Exercise 4

a. The Main Entities

- **Customer:** Individuals who purchase items
- **Comic_Book :** The physical print items available in inventory
- **Order:** The customer's purchase.
- **Creator:** All creators contributing in the issue (the comic): The writers, colorists ... etc
- **Publisher:** The company that prints and distributes the comic books

c. The Relationships Between These Entities

- **A Customer can place one or more Orders**, while an order is linked to exactly one paying customer.
- **An Order can contain several Comic Books**, and a specific comic book can be sold across multiple customer orders.
- **A Comic Book is worked on by several Creators**, and an individual creator can produce work across many different comic books.
- **A Comic Book belongs to exactly one Publisher**, and a publisher distributes many different comic books.